

Campus parking and exit carpool coordination

WORKSHOP 3: Robust System Design and Project Management

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1. Executive Summary

This document presents the enhanced conceptual design and project planning framework for the Campus Parking and Exit Carpool Coordination, an information system designed to manage capacity and achieve greater parking efficiency at the engineering faculty of Universidad Distrital, as well as coordinate shared trips from the campus to key zones across the city. The project builds on Workshops 1 and 2, which evidenced severe parking saturation through data collected from a survey distributed to students, professors, and staff who utilize the parking facilities. This data reveals a daily overflow of approximately 90 motorcycles onto surrounding streets, access bottlenecks at the Calle 40 basement due to traditional management methods, and a heavy reliance on informal coordination through messaging apps.

The initial proposal was originally framed as Sustainable Transportation Coordination, combining inter-campus carpooling and a bike-sharing system. However, a feasibility analysis revealed that an institutional bike-sharing program is neither realistic nor feasible given infrastructure, safety, and operational constraints. Consequently, the project scope was refined and the system was renamed Campus Parking and Exit Carpool Coordinator, shifting its focus toward system design for parking inventory and reservation management, policy-based prioritization for shared vehicles, and exit-time carpool matching toward different city zones, the platform is only one enabling component inside a broader operational system involving institutional rules, physical parking resources, security procedures, and stakeholder coordination.

Workshop 3 does not involve software implementation; instead, it delivers a robust conceptual architecture, a risk and quality framework aligned with ISO 31000 and ISO/IEC 25010 standards, and a project management plan to guide future development efforts. This

framework analyzes factors from multiple angles using a stakeholder role-playing approach, evaluating all viewpoints to contribute to an integrated development that encompasses human, organizational, and structural factors. Through these interactions and a holistic approach, realistic behavioral patterns are strengthened, ensuring that the potential actions of the system are fully accounted for to achieve optimal development.

2. System Evolution

Workshop 1 – System Analysis

The team analyzed transportation and parking conditions and factors at the engineering campuses using field observations, parking occupancy measurements, process documentation, and a survey conducted among 51 users. The focus was to assess the infrastructure while capturing issues reported by respondents to understand user behavioral patterns. The results showed that 55% of respondents arrive before 8:00 a.m., marking a recurrent saturation in the Calle 40 basement, an overflow of approximately 90 motorcycles onto public streets, and access control based on a mix of digital registration and manual decisions by security personnel. Users rely heavily on public transport and informal carpooling coordinated through messaging apps within close social circles, which motivated the need for institutional coordination mechanisms.

Workshop 2 – Initial Architecture

Based on these findings, Workshop 2 defined a dual-focused structure that included an application and a security dashboard, core services for ride matching and status updates, an integration layer for vehicle registration and user authentication, and a data and monitoring layer for trip and parking records. This preliminary design still reflected the original Sustainable Transportation Coordinator concept (inter-campus trips and potential

bike-sharing support), but already incorporated scalability, reliability, and data privacy requirements for future operation.

Scope Refinement and Workshop 3

Between Workshops 2 and 3, the team reassessed the feasibility of the original concept and concluded that the project needed a more realistic and operationally viable focus. From the perspective of systems analysis and design, feasibility is a central factor in project definition, because even a technically functional solution loses value if it cannot be applied under real conditions. In this case, the initial approach presented important limitations due to changes in the institutional context, particularly the fact that the contract with the ECCI headquarters is ending. Under this new scenario, the original emphasis on broader campus carpooling became less relevant, since many students would no longer face the same mobility conditions that originally justified that part of the proposal.

For this reason, the team decided to reformulate the project scope and concentrate on the aspects that remain both urgent and feasible. The short-term implementation of a bike-sharing system was discarded, and the broader inter-campus mobility logic was reduced. In its place, the project was reoriented toward parking space management and exit-time carpool coordination, focusing on the reservation, allocation, supervision, and release of parking spaces, while also supporting shared vehicle departures from the university toward students' destination zones across the city.

As a result of this refinement, the project was renamed Campus Parking and Exit Carpool Coordinator, and its boundaries were restructured accordingly. Workshop 3 therefore no longer focuses on describing a future software product in isolation, but on defining a robust conceptual system that integrates operational rules, stakeholder roles, physical constraints, risk management, quality attributes, and implementation planning. This shift also

reflects a more mature systems-engineering perspective, because the project is now understood as a socio-technical coordination system rather than as a simple digital application.

2.1 Stakeholder Analysis

Methodological Approach

The stakeholder analysis is conducted within the identification, classification, and impact framework proposed by systems thinking: each stakeholder is characterized by their role within the system, their functional needs, their mode of interaction with the system, and the benefits they derive. Furthermore, potential frictions that could affect the adoption or operation of the system are identified.

Stakeholder table

Stakeholders	Role in the system	Needs	Interaction	Gains	Potential frictions
Students	Primary user	Reserve parking, find carpooling, receive notifications	Reservation compliance, arrival behavior, exit coordination, and response to institutional rules	Time savings, guaranteed space, reduced transportation costs, convenience	Resistance to change. Informal WhatsApp.
Security Guards	Field Operator	Real-time view of the parking lot, access control, and	The operational bridge between digital	Less chaos at the entrance, digital support for decisions	Technological learning curve

		alerts for anomalies	coordination and physical access control.		
Faculty and staff	Secondary user with vehicle	Reserve spaces according to class schedule, availability visibility	App/web: same functions as students, possible differentiated priority	Spaces assigned in advance, less time spent looking for parking	They may perceive that the system prioritizes students
Student Welfare	Influential Stakeholder	Student mobility data, support for adoption campaigns	Indirect: receives system reports, co-leads institutional campaigns	Quantitative evidence of impact on student quality of life	May not prioritize the project if there is no institutional alignment
Campus Administration	Operational Authority	Parking lot management, policy enforcement, violation reporting	Administrative Dashboard: reports, policy configuration, auditing	Parking lot conflict reduction, incident tracking	Resistance if the system requires changes to current processes
IT Department	Technical Enabler	Integration with @udistrital.edu.co, data security, stable infrastructure	Backstage: manages institutional authentication, servers, and permissions	Standardization of access to university systems	Additional workload of technical support without allocated resources
Environmental management units	Indirect benefit stakeholder	Data on reduction of individual vehicles, environmental metrics	Indirect: consumes system reports for institutional reporting	Evidence of positive environmental impact (reduced CO ₂ emissions)	Lower if there is no formal communication channel with the project

Stakeholders do not interact only with interfaces, but with policies, physical constraints, and decision procedures, which is why the system must be designed beyond software boundaries.

University Welfare

University Welfare does not interact directly with the system during its daily operation, but it plays a strategic role in its adoption. This unit requires quantitative evidence regarding student mobility conditions to support its support programs. The system provides real-time data on parking usage patterns, carpooling frequency, and the most common destination zones, enriching their understanding of the students' living context. Their primary benefit is having access to information that was previously invisible or fragmented across WhatsApp groups, alongside the possibility of providing data and information for system feedback. The potential friction lies in the fact that Welfare may not perceive the system as part of its institutional mandate; therefore, it is necessary to establish a formal reporting channel and align the project with their quality-of-life objectives. Within the implementation strategy, Welfare is the natural ally for the adoption campaign described in Section 7.2.

Campus Administration

Campus Administration is the authority that defines and enforces parking usage policies. Their core need is traceability: knowing who accessed the facilities, when, with which vehicle, and whether the usage regulations were respected. Currently, this information exists in a fragmented state between manual logs and arbitrary decisions by security guards. The system offers them an administrative dashboard featuring audit logs, violation alerts, and the capability to configure prioritization rules (for instance, giving preference to vehicles with more than two occupants). Their main benefit is transitioning from reactive management to data-driven management. The most significant friction is that the administration may perceive the system as a threat to established processes or as an additional supervisory burden; consequently, their involvement from the design stages is critical to ensure institutional legitimacy.

Faculty and Staff

Their core need is to secure a parking space based on their teaching or work schedules, which differ from those of students and are generally more predictable. They interact with the system through the same web/mobile interface as students, although the Parking Allocation and Policy Engine could incorporate differentiated rules based on institutional roles. Their primary benefit is eliminating the uncertainty of arriving and not finding an available space. The potential friction is that if the system does not explicitly account for their usage patterns (such as different arrival times or larger vehicle sizes in some cases), they may feel the design is exclusively student-oriented, which would reduce adoption and generate conflict with security guards at access points.

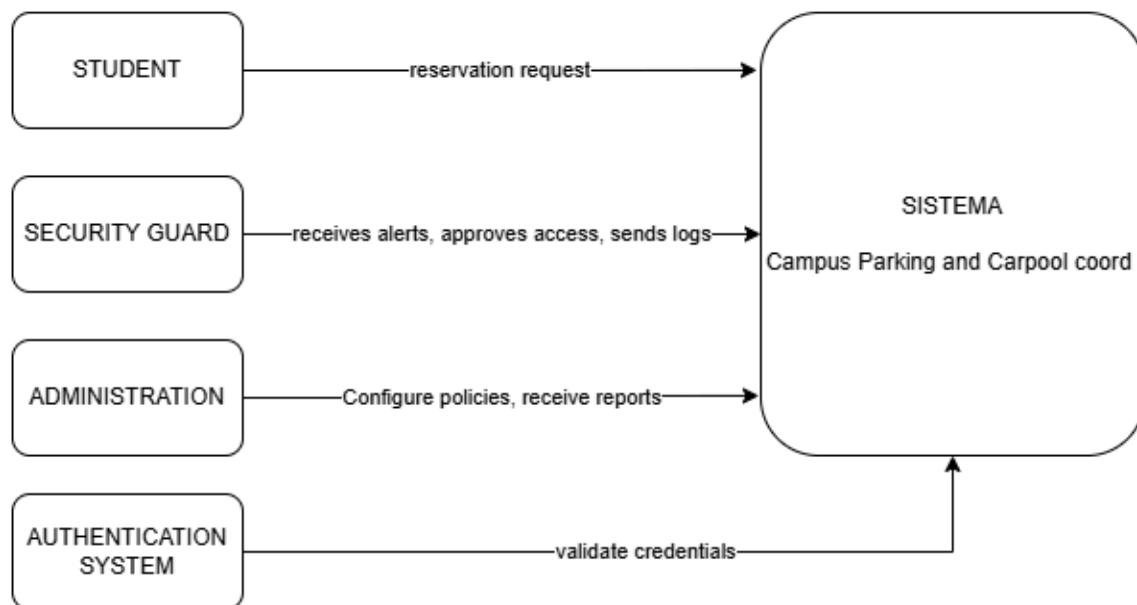
2.2 Problem Modeling

The current situation and the expected transformation of the system are modeled. This section uses three complementary visual models to characterize the problem space: a context diagram that delimits the system boundaries and its external actors, an As-Is vs. To-Be process flow that contrasts current informal practices with the proposed coordinated workflow, and an Input-Process-Output (IPO) model that defines what the system receives, transforms, and produces. Together, these models provide the analytical foundation that justifies the architectural decisions presented in Section 3.

2.2.1 Context Diagram

The context diagram establishes the boundary of the Campus Parking and Exit Carpool Coordinator system by identifying the external entities that interact with it and the nature of those interactions. Rather than describing internal components, this model focuses on what

flows in and out of the system from the perspective of each actor, making explicit which stakeholders are within the system's operational scope and which remain external to it.

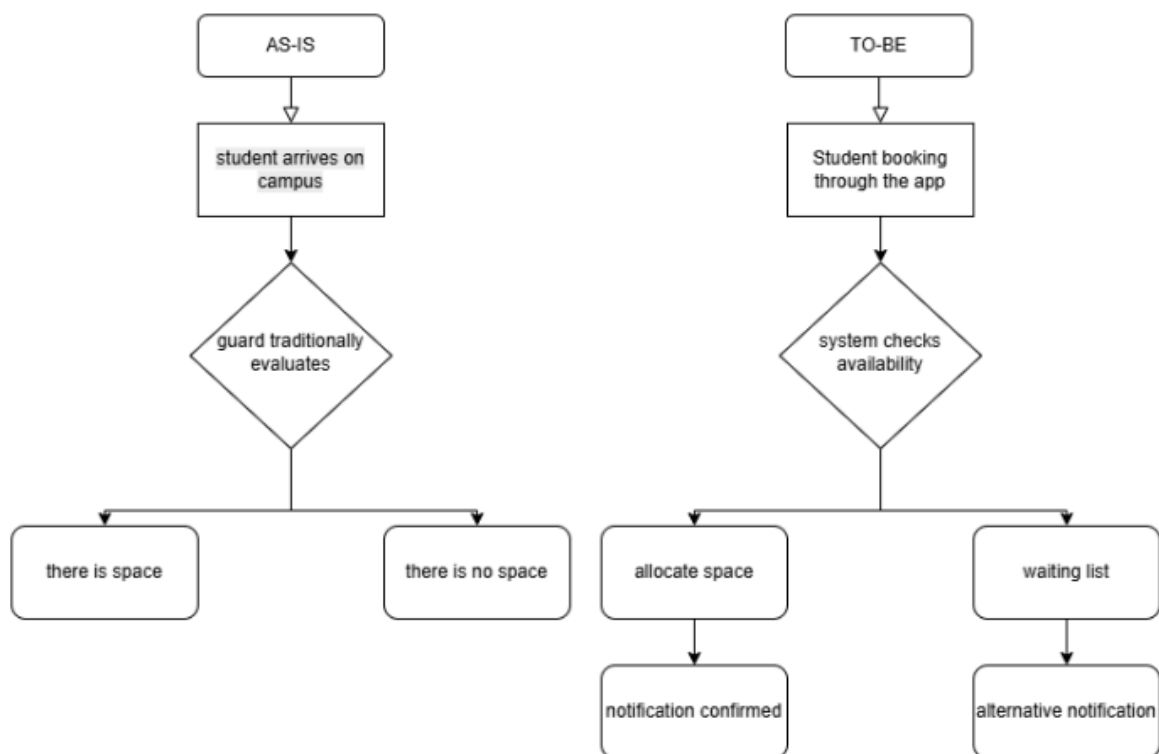


The diagram reveals four primary external actors. Students are the main users of the system, initiating reservation requests for parking spaces and carpool matching; their interaction is direct and frequent, occurring at peak entry and exit windows. Security guards occupy a dual role: they both receive operational information from the system —such as real-time occupancy alerts and access approvals— and feed data back into it through entry and exit logs, making them a critical link between the digital system and physical campus access control. The campus administration interacts with the system at a policy and oversight level, configuring prioritization rules and receiving aggregated reports for institutional decision-making. Finally, the institutional authentication system of Universidad Distrital functions as an external validator, planned to ensure in a future phase that only users with verified @udistrital.edu.co credentials can access the platform. For this current conceptual

phase, a standardized user registry is proposed, leaving the foundation prepared for future institutional email integration.

2.2.2 As-Is vs. To-Be Process Flow

The following diagram contrasts the current inefficient process for parking access with the coordinated workflow proposed by the system. The As-Is flow represents the situation documented during the Workshop 1 field observations, while the To-Be flow reflects the conceptual design developed across Workshops 2 and 3.

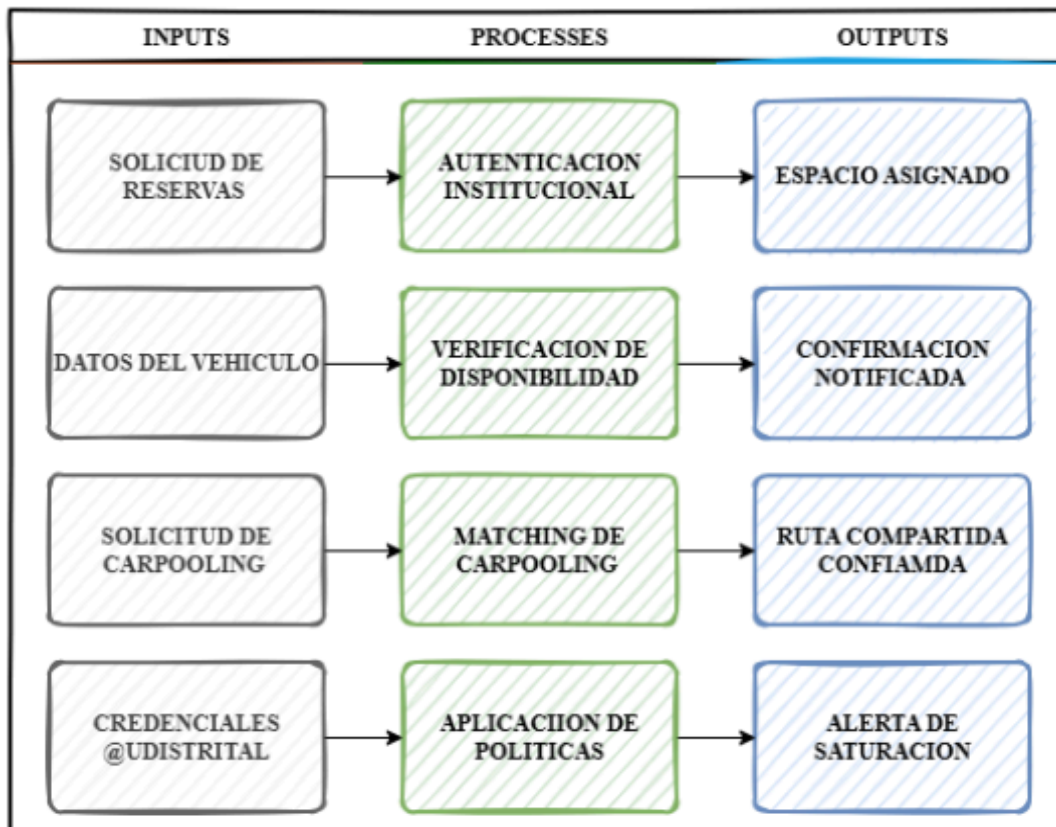


The As-Is flow exposes two fundamental weaknesses in the current process. First, the access decision depends entirely on a visual evaluation by the security guard, introducing subjectivity, inconsistency, and a single point of failure at the Calle 40 entrance. Second, when no space is available, the process has no structured outcome: the student is implicitly

redirected to park on surrounding public streets, generating the overflow of approximately 90 motorcycles documented in Workshop 1, without any record or notification being generated. The To-Be flow addresses both failure points systematically. The student interacts with the system before arriving on campus, shifting the coordination moment from physical arrival to prior digital reservation. The availability check becomes an automated, data-driven decision rather than a human judgment call, and both outcomes —space allocation and waiting list placement— generate notifications that keep the user informed regardless of the result. This transition from reactive, manual control to proactive, system-mediated coordination is the central operational improvement the proposed system delivers.

2.2.3 Input-Process-Output (IPO) Model

The Input-Process-Output model defines the informational transformation logic of the system at a conceptual level. It specifies what data the system receives from its actors, the core processes through which that data is transformed, and the operational outcomes produced for users and administrators. This model complements the context diagram by shifting focus from who interacts with the system to what the system does with the information it receives.



The model identifies four input-process-output chains that together cover the complete functional scope of the system. The reservation request, combined with institutional credential validation through the authentication process, produces an assigned parking space—the primary outcome users expect from the system. Vehicle data feeds into the availability verification process, generating a confirmation notification that closes the reservation loop for the student. Carpool requests are transformed through the matching process into confirmed shared routes, directly replacing the informal coordination currently handled through messaging applications. Finally, institutional credentials serve as input to the policy application process, which produces saturation alerts that enable security staff and administrators to respond to capacity thresholds before they generate physical overflow. Taken together, these four chains demonstrate that the system's core value lies not in storing

data, but in transforming fragmented, informal inputs into structured, actionable outputs for every actor in the system.

2.3 Survey Data Integration

The following section presents the key findings from the structured survey conducted during Workshop 1 as part of the data collection phase. The survey was administered to 51 members of the engineering faculty community at the Calle 40 campus and covered user role, frequency of campus visits, dominant transport mode, arrival time, travel time, parking availability perception, shared travel practices, and interest in institutional coordination tools. This data is integrated into the Workshop 3 design to demonstrate that the architectural decisions and system scope are grounded in empirical evidence rather than assumptions.

2.3.1 Survey Questions and Purpose

The survey tool was structured around six core variables, each designed to inform a specific dimension of the system design:

Question	Purpose	Design implication
User role (student / staff)	Identify the primary user base	Confirms a student-centered interface design
Dominant transport mode	Understand current mobility patterns	Justifies carpooling as a priority
Most common arrival time	Identify peak demand windows	Defines reservation and matching time slots
Parking availability perception	Measure saturation experience	Validates parking management as an urgent need

Willingness to use carpooling	Gauge adoption potential	Informs incentive strategy design
Willingness to use bike-sharing	Gauge adoption potential	Informed scope refinement decision

2.3.2 Key Results and Analysis

The survey produced five headline findings, each of which directly shaped a decision in the Workshop 3 design:

- **Finding 1** — The surveyed user base is almost entirely students (98%). The near-total dominance of students as system users confirms that the interface design must be optimized for a student context: mobile-first, and designed to support a future integration with institutional credential login (@udistrital.edu.co) and academic schedules. The inclusion of faculty and administrative staff as secondary users in the stakeholder analysis does not contradict this finding; rather, it reflects that, although their volume is low, their parking priority and behavioral patterns differ enough to warrant differentiated Parking Allocation policy rules.
- **Finding 2** — Public transport is the dominant mode (60.8%), but informal carpooling already exists. The majority of users do not arrive by private vehicle, which might seem to reduce the urgency of a carpooling system. However, the process documentation from Workshop 1 confirmed that informal ride-sharing coordination through WhatsApp and Facebook groups is already widespread, driven by cost and safety motivations rather than parking availability. This indicates that the community has reached a level of sociotechnical readiness for a formal coordination platform: the behavior exists, but lacks structure and institutional support. The system does not need to create a new habit; it needs to formalize one that already works informally.
- **Finding 3** — 54.9% of users arrive before 8:00 a.m. This is the most operationally significant finding for the system design. More than half of campus users arrive in a single early-morning window, which creates a predictable and recurring saturation peak at the Calle 40 entrance. This directly justifies the definition of peak departure and arrival windows in the Exit Carpool Matching Service, and informs the Performance Efficiency metric established in Section 5: the system must handle up to 500 simultaneous requests precisely during the 7:00 to 8:00 a.m. and 12:00 to 2:00 p.m. windows.

- **Finding 4** — Only 34.3% expressed willingness to use carpooling This relatively low adoption intention for carpooling is a critical design constraint that the risk management framework directly addresses through Risk P5. This reinforces the need for an interface where publishing a carpool slot takes fewer than three interactions, and supports the incentive strategy of offering parking priority to active carpool contributors. Parking management and carpool coordination must function as complementary modules, not as a single solution.
- **Finding 5** — 74.5% expressed willingness to use bike-sharing The high interest in bike-sharing initially made it the centerpiece of the Workshop 1 and 2 design. However, the scope refinement process conducted between previous Workshops concluded that an institutional bike-sharing system is not viable in the short term due to infrastructure, safety, and operational constraints at the Calle 40 campus. Therefore, this finding serves a different purpose in Workshop 3: it confirms that there is latent demand for lower-emission mobility alternatives, which positions the carpooling system as a first step in a longer sustainability trajectory. The bike-sharing component is not discarded permanently; it is deferred to a future implementation phase.

2.3.3 Connection Between Survey Evidence and Design Decisions

The following table makes explicit how each survey finding maps to a specific component or decision in the design, justifying architectural choices with empirical data:

Survey finding	System component justified
98% of students, institutional email required	Authentication Service: designed with a scalable framework to incorporate @udistrital.edu.co restriction in a future phase.
Informal carpooling via WhatsApp already exists	Exit Carpool Matching Service: replaces informal coordination
54.9% arrive before 8:00 a.m.	Performance metric: peak window; reservation time slots
34.3% willing to use carpooling	Risk P5 mitigation: 3-click interface + parking priority incentive

74.5% willing to use bike-sharing	Scope refinement: deferred to a future phase, not discarded
Daily overflow of ~90 motorcycles to public streets	Parking Inventory Service: real-time occupancy as a core function

3. Architecture Refinement

3.1 Conceptual Architecture Overview

The conceptual architecture of the Campus Parking and Exit Carpool Coordinator is organized as a socio-technical system that connects students, security staff, campus administration, institutional rules, physical parking capacity, and shared exit coordination. At this level, the system is understood as a coordinated whole whose behavior emerges from the interaction between actors, rules, information flows, and operational decisions, rather than as a software product defined only by implementation artifacts.

The architecture includes four main functional components: student participation, parking state management, exit coordination, and operational control. The student participation component introduces requests, updates, and coordination intentions into the system; the parking state management component represents and updates the condition of available, reserved, occupied, blocked, and priority spaces; the exit coordination component relates drivers and passengers according to compatible departure conditions; and the operational control component supervises compliance, exceptions, and intervention needs. These components are linked through flows of information and control: student actions affect parking and coordination states, parking availability constrains allocation decisions, matching

outcomes influence later departures, and operational supervision provides corrective feedback when misuse, inconsistency, or exceptional conditions arise.

This architecture is intentionally conceptual. Its purpose is to define the responsibilities and relationships that explain how the system behaves as a coherent operational arrangement, not to commit to a specific software stack or interface design. In that sense, the platform is only one supporting element within a broader system that also includes institutional policies, guard procedures, user behavior, and the physical parking environment.

3.2 Conceptual Interaction Structure

The system defines two main conceptual interaction spaces. The first is the student participation space, where users interact with the system to consult parking availability, request or release parking spaces, register vehicle information, announce departure intentions, and coordinate shared exits according to destination zones and time compatibility. This interaction space is not described as a specific application screen, but as the set of exchanges through which students provide information to the system and receive allocation or coordination outcomes.

The second is the operational control space, which groups the interactions required by guards and university operational actors to supervise parking occupancy, verify access conditions, review active reservations, detect misuse situations, and intervene in exceptional cases. At the conceptual level, this space represents a control and monitoring function within the system rather than a concrete dashboard. It connects the digital coordination process with physical access control, which is essential in a real campus parking environment.

These interaction spaces are conditioned by the actual capacity of the parking area, institutional schedules, peak entry and exit times, and the need for manual intervention in

exceptional situations. For that reason, the design remains focused on information flows, responsibilities, and decision support, which is consistent with the workshop's conceptual scope.

3.3 Conceptual Functional Subsystems

At the conceptual level, the system is organized into functional subsystems with differentiated responsibilities. The access control subsystem governs identity validation and permission recognition. The parking state subsystem represents the condition of spaces as available, reserved, occupied, blocked, or priority. The allocation and policy subsystem applies the institutional rules that define who can access a space, under what conditions, and with what priority.

In parallel, the exit coordination subsystem manages the matching of drivers and passengers according to destination compatibility, departure windows, and seat availability, while the information update subsystem ensures that relevant state changes are communicated to the corresponding actors. This decomposition is useful because it clarifies the internal logic of the system as a set of coordinated responsibilities and feedback relations, not as a commitment to a specific implementation technology.

The main purpose of this decomposition is to keep the system understandable and traceable. Each subsystem responds to a concrete operational need: institutional legitimacy, space consistency, policy enforcement, shared departure coordination, and timely notification. Therefore, the design remains within the intended level of abstraction for Workshop 3, where the emphasis is on system organization, behavior, and robustness rather than on software construction.

3.4 Conceptual Processes and Information Flows

The system behavior can be understood through a set of conceptual processes that transform user inputs into coordinated operational outcomes. In parking coordination, the process begins when a student communicates the need to access or release a parking space;

this input is evaluated according to current parking state and institutional allocation rules, producing an updated system state and a resulting decision such as assignment, denial, priority handling, or exception escalation.

In exit carpool coordination, the process begins when a driver communicates departure intention and available capacity, or when a passenger expresses a compatible travel need. The system then relates these inputs according to destination compatibility, departure window, and seat availability, generating either a successful coordination result or a pending or unresolved state if no suitable match exists. This process is conceptually important because it shows how the system connects independent actors through shared timing and mobility conditions rather than through isolated actions.

The model also includes supervisory and feedback processes. Control actors observe parking occupancy, detect irregularities or exceptional situations, and intervene when necessary to preserve order, continuity, and consistency. These feedback actions help maintain system equilibrium, which is consistent with the systems-thinking perspective that emphasizes regulation, interconnectedness, and homeostasis as core properties of viable systems. When a request cannot be fulfilled, the system should not simply fail; it should produce an understandable outcome and preserve traceability for later review.

3.5 Design Justification and Boundaries

Each architectural element responds to a specific operational need in the current project. Access control exists to ensure institutional legitimacy and prevent unauthorized use. Parking state management exists to keep occupancy consistent and reduce disputes about available spaces. Allocation and policy exist to make decision-making explicit instead of leaving it to informal judgment. Exit coordination exists to replace fragmented WhatsApp-based coordination with a structured shared-departure process. Operational

control exists to handle exceptions and preserve order in a physical environment where digital coordination alone is not enough.

The system boundary includes the parking area, the student and staff actors, the security personnel, the institutional policies, and the information flows needed to coordinate access and shared exits. External dependencies include the university's authentication service, broader campus rules, and environmental conditions such as peak-hour congestion and physical limitations of the parking lot. The public street is not part of the system, but it remains an external condition that reflects the consequences of poor coordination.

This distinction is important because the project is not simply a platform for reservations. It is a coordination system that uses software as a support mechanism for a broader operational process. Keeping that boundary clear helps avoid reducing the design to a web application and preserves the systems-engineering perspective required by the workshop.

4. Risk Management

The design must account for both technical and human dimensions, since the system operates as a socio-technical arrangement. Its robustness depends on the interaction between parking rules, matching logic, operational control, institutional policy, and user behavior. For that reason, this section combines risk analysis with stakeholder considerations, giving balanced attention to operational, technical, institutional, and human factors.

4.1 Risk identification

The first step is to identify the main risks that may affect the functioning of the system. One relevant risk is low user adoption, especially if students continue preferring informal coordination channels instead of a formal institutional system. Another risk is cancellation, no-show, or repeated misuse behavior, since these actions can reduce trust, weaken shared travel coordination, and undermine the credibility of reservation-based parking management.

A technical risk is incorrect or delayed state synchronization between parking availability, reservations, and actual occupancy, which could produce inconsistencies in the system. Other risks include congestion during peak hours, privacy and security issues related to mobility and access data, and institutional misalignment if campus administration, student welfare, and security personnel do not share clear procedures and responsibilities. Human factors are also relevant, because resistance to change, low digital confidence, or misunderstanding of the rules may limit effective use of the system.

4.2 Risk Classification

Once identified, the risks should be classified according to their nature. Operational risks include low adoption, cancellations, no-shows, misuse of reservations or priority spaces, and congestion during peak periods. Technical risks include state mismatch, failure in reservation logic, poor system responsiveness, or loss of data consistency.

Institutional risks include lack of policy alignment, unclear responsibility among stakeholders, or limited administrative support. Security and privacy risks include unauthorized access to personal or mobility-related information, misuse of parking privileges, or exposure of sensitive user data. Human-related risks include mistrust, low adoption, lack of training, preference for informal coordination, and loss of confidence caused by repeated unfair or unreliable behavior within the system.

4.3 Risk Impact on the System

Each risk may affect the system differently depending on its frequency and severity. Low adoption would reduce the usefulness of the system because the value of carpooling and coordinated parking depends on having enough active participants. Cancellation and no-show behavior could damage trust and reduce the reliability of the coordination process. If parking state updates are delayed or inconsistent, the system may make incorrect allocation decisions, which would directly affect credibility and operational efficiency.

Congestion during peak hours could create stress on the system and on the people operating it, especially guards and campus staff. Privacy or security incidents could seriously damage user confidence and make students reluctant to register vehicles or share mobility information. Institutional misalignment could slow down implementation or create inconsistent rules across stakeholders. Human factors may also affect impact indirectly: even a technically correct system may underperform if users do not understand it, do not trust it, or do not see enough value in changing their habits.

4.4 Conceptual Mitigation Strategies

To reduce these risks, the system should include conceptual mitigation strategies rather than implementation details. For adoption-related risk, the university should support onboarding actions, clear communication, and institutional endorsement so that students understand the purpose and benefits of the system. For cancellation and no-show behavior, the conceptual design should include reservation accountability, visibility of participation history, and rule-based consequences for repeated misuse.

For technical synchronization risks, the system should be designed with state verification, periodic consistency checks, and clear separation between requested, reserved, and confirmed states. For congestion at peak hours, the system can prioritize requests according to predefined institutional rules and include a controlled allocation process for high-demand periods. For privacy and security risks, access should be limited according to role and need, and sensitive information should only be visible to authorized actors.

For institutional risks, the main mitigation is governance: responsibilities must be clearly assigned among student services, security personnel, and campus administration. For human-factor risks, the system should be easy to understand, transparent in its rules, and consistent in its behavior so users can build trust over time. The goal is not to eliminate all

risk, but to make the system resilient enough to continue operating under realistic academic conditions.

4.5 Participation Accountability and Trust Mechanisms

Because the proposed system depends on repeated coordination among independent users, trust and accountability must be treated as operational stability conditions rather than as optional digital features. In the current informal context, students often rely on personal familiarity or small messaging groups to judge whether a parking arrangement or shared exit is reliable. Once coordination becomes institutional, the system must provide clear mechanisms to preserve trust, discourage misuse, and make participation rules visible to all actors.

Several behaviors may weaken confidence in the system, including repeated late cancellations of carpool offers, no-show after reserving a parking space, misuse of priority spaces, inaccurate vehicle or departure information, and recurrent disregard of the operating rules. Although each event may seem minor on its own, their accumulation can reduce user confidence, increase manual intervention by guards, and weaken the reliability of both parking allocation and exit coordination.

To reduce this risk, the conceptual design should include a participation accountability mechanism based on traceable operational records. The system should register relevant events such as reservation requests, confirmations, cancellations, arrival validation, repeated no-show cases, manual overrides by guards, and reports of misuse. These records are not intended as punitive surveillance, but as an operational memory that allows the institution to distinguish isolated exceptions from repeated patterns that affect system stability.

Consequences should be gradual, transparent, and proportionate. An isolated incident may only generate a warning or reminder, while repeated no-show behavior or misuse of reservation privileges may temporarily reduce booking priority, restrict advance reservations,

or trigger administrative review. This logic is important because it protects fairness without over-penalizing occasional justified situations.

To avoid arbitrariness, the rules must be explicit. Users should know which behaviors affect their participation status, how these events are recorded, who can review them, and under what conditions any restriction may be applied or removed. In the same way, guards and administrators should intervene according to defined criteria, so that exceptions remain auditable and consistent with the institutional policy framework of the system.

From a systems perspective, this mechanism does more than regulate behavior. It strengthens adoption, improves the credibility of reservations and carpool matching, reduces avoidable conflicts, and provides evidence for future adjustments of the operating rules. In this way, trust becomes a design condition supported by transparency, traceability, and coordinated institutional oversight.

5. Implementation Strategy

5.1 Implementation Approach

The implementation strategy for our parking management system will follow a phased and controlled approach, because the system must coordinate access, reservations, availability, and vehicle movement in a real academic environment. The goal is to move from the conceptual design produced in the previous workshops to an operational solution that can support daily parking decisions without causing confusion or conflict among users. Since the system depends on multiple actors and rules of use, it is better to introduce it gradually rather than deploy it all at once.

5.2 Institutional Preparation

Before the system is deployed, the parking rules of the institution must be clearly established and validated. This includes defining who can reserve a space, how priorities are assigned, what happens when a user does not arrive on time, and how exit coordination is

managed when a blocked vehicle must leave first. In our project, this step is essential because the system is not only technical; it also organizes behavior, responsibilities, and decision-making inside the parking area.

5.3 Pilot Deployment

The first real use of the system should be done through a pilot with a reduced group of users and a limited number of parking spaces. This pilot will allow us to test whether the reservation logic, the availability control, and the exit coordination rules work correctly under realistic conditions. It also helps us observe user behavior, detect misunderstandings in the process, and identify whether the design supports the daily flow of the parking lot as expected.

5.4 Validation of Operating Rules

During the pilot, the system must be validated against the main scenarios of our project: normal parking allocation, reservation confirmation, late arrival, blocked exit, and conflict resolution between users. The objective is to verify that the system behaves according to the requirements defined in the analysis and design stages. If the system fails to reflect the actual operating rules of the parking area, then the design must be adjusted before broader adoption.

5.5 Controlled Expansion and Change Management

If the pilot is successful, the system can be expanded gradually to a larger number of users and parking spaces, but this growth must be managed as a controlled system evolution rather than as a sudden release. The reason is that the value of the proposed solution depends on the stable interaction between parking availability, reservation behavior, exit-time carpool coordination, security oversight, and institutional policy enforcement; if any of these elements scales faster than the others, the system may become inconsistent or difficult to operate.

The expansion should therefore follow a staged process. First, the number of active users should increase within the same physical area and operating rules. Only after that stage is stable should the system extend to additional user groups, more parking spaces, or broader departure-zone coordination. This approach allows the team to observe whether the rules remain understandable, whether guards can maintain control at peak hours, and whether the coordination logic continues to work without creating congestion or confusion.

During expansion, the team should monitor a small set of operational indicators to detect bottlenecks early. These indicators should include the volume of reservation requests during peak periods, the percentage of no-shows or late arrivals, the number of conflicts resolved manually by security staff, the average time needed to confirm or reject a parking request, and the frequency of unresolved carpool matches. If any of these indicators rises beyond an acceptable threshold, the scope of the rollout should be paused and the operating rules should be adjusted before continuing.

Change management is also essential because the system alters daily routines and decision-making. Any change in reservation rules, priority criteria, blocked-space handling, or exit coordination must be documented and communicated in advance to students, faculty, security personnel, and campus administration. This is especially important because the system does not replace the institutional rules; instead, it formalizes them and makes them traceable, so users need to understand not only how to interact with the system but also why the rules exist and how exceptions will be handled.

From a systems perspective, expansion should preserve the system boundaries and responsibilities already defined in the conceptual model. That means the software component should continue serving as a coordination support tool, while the physical parking lot, the security team, the institutional authentication service, and the university policy framework remain part of the broader operational system. In this way, growth does not mean adding

features indiscriminately; it means increasing capacity and coverage while keeping the system understandable, auditable, and aligned with the university's operational reality.

5.6 Training and Operational Readiness

The system will only be ready for full use when users and administrators understand how it works and how to respond to exceptions. For our project, this means preparing simple instructions for parking users, defining how administrators handle conflicts, and establishing what to do when the system cannot resolve a situation automatically. Readiness also requires that the team has monitoring procedures, support responsibilities, and a clear way to incorporate feedback for future improvements.

6. Design Validation and Final Recommendations

Validation of the proposed design must confirm that the system responds not only to the technical expectations, but also to the operational conditions identified for the project solution. In this case, validation means verifying whether the conceptual design is capable of coordinating parking access, maintaining consistency in parking management, supporting the coincidence of shared exits, and remaining understandable to users, guards, and administrators. This is aligned with the workshop's goal of preparing the system for implementation rather than limiting the analysis to building the software.

From a systems perspective, the current design shows clear progress compared to previous workshops. The project now defines a broader operational boundary that includes stakeholders, institutional rules, physical parking capacity, oversight mechanisms, and information flows, rather than focusing solely on the software platform. This improvement is important because it addresses the need to understand the solution as a sociotechnical system whose performance depends on the interaction between people, rules, and technical support mechanisms.

The architecture can be considered conceptually valid because it organizes the system into coherent functional responsibilities: identity and access control, parking status management, policy assignment and enforcement, egress time coordination, and operational oversight. These are connected through information flows and decision rules that explain how requests are transformed into operational results. As a result, the architecture now reflects system behavior more clearly and reduces the risk of interpreting the project as only a web application structure.

The design is also strengthened by the inclusion of explicit risk thinking. Low adoption, cancellation behavior, no-show, misuse of reservations, congestion, institutional misalignment, and synchronization problems are already recognized as relevant threats. The addition of accountability and trust mechanisms improves this framework because it connects user behavior with rule-based consequences, operational traceability, and clearer institutional oversight, which are all necessary to sustain reliability in real use conditions.

Even so, some recommendations remain important before treating the design as fully mature. First, the operational rules should be made even more explicit in the future pilot stage, especially regarding late arrivals, blocked exits, priority allocation, and repeated misuse. Second, the process logic could be represented with greater precision through BPMN-style models for critical scenarios, particularly where multiple actors, exceptions, and manual interventions are involved. Third, the monitoring framework should include concrete indicators such as no-show frequency, unresolved carpool matches, parking state mismatch rate, and average intervention time for exceptions, so that the system can be evaluated with observable evidence after deployment.

A further recommendation is to preserve the distinction between the broader system and the software platform in all future iterations. The platform should continue to be presented as a support mechanism for institutional coordination, not as the system itself. This

is important because the effectiveness of the proposal ultimately depends on user behavior, guard procedures, organizational legitimacy, and physical capacity constraints, none of which can be solved by interface design alone.

In validation terms, the design can therefore be considered consistent with the objectives of Workshop 3. It demonstrates evolution from problem analysis to conceptual system organization, and from initial architecture to a more robust framework that includes risk, quality, change management, and implementation readiness. The main value of the current version is that it no longer treats the project as an isolated digital product, but as an operational coordination system intended to function under realistic university conditions.

7. Conclusión

The evolution of this project shows that a systems design and analysis approach only makes sense when technical ideas are tested in real operating conditions, which is why it was necessary to change the direction of our project. What began as a broader mobility proposal was refined into a more focused and viable sociotechnical system after recognizing the limitations of the original scope and the changing institutional context.

The main contribution of the project is not only the definition of a parking and ridesharing coordination platform, but also the identification of the elements of the system that really determine its success: stakeholders, rules, physical limitations, trust, responsibility and operational control. By modeling these elements together, the project goes beyond a software-centric view and treats parking management as a coordinated institutional process.

The conceptual architecture developed in Workshop 3 clarifies that the system must support parking assignment, departure time matching, and exception monitoring through a constant flow of information and responsibilities. This makes the proposal more robust because it connects the needs of users, administrators, and campus operations within a design that can be understood, validated, and improved over time.

Risk management also became a central part of the design. Issues such as low adoption, no-show behavior, misuse of reserves, timing errors, and institutional misalignment were not treated as secondary problems, but as conditions that directly affect the stability of the system. In that sense, the project demonstrates that a viable design must include accountability and trust mechanisms, not just functional features.

Finally, the project demonstrates that feasibility is a design principle, not a final verification. The change in scope allowed the team to focus on what is truly implementable and useful for the university community in the current scenario. This refinement did not weaken the project; On the contrary, it made the proposal more realistic, more coherent, and better aligned with the needs of the campus environment.

8. References

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